Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [14EC44]

FOURTH SEMESTER B.TECH UNIVERSITY EXAMINATION, JUNE – 2016

SIGNALS AND SYSTEMS

Time: 3 Hours **Maximum Marks:70**

Instructions:1. **Answer any FIVE full questions by choosing at least 2 Questions from each part**

2. **Any missing data may be suitably assumed**

PART - A

1. a. Sketch the following signals. **04**

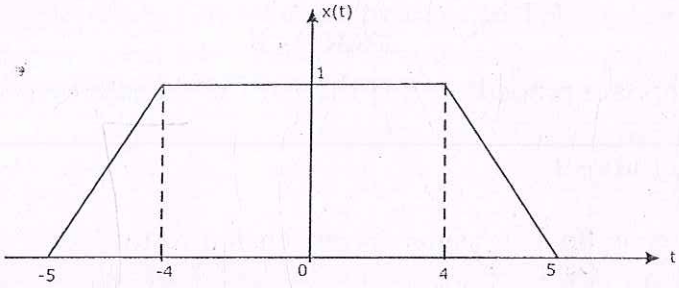
i. $\pi\left(\frac{t-1}{2}\right)+\pi(t-1)$

ii. $r(-0.5t+2)$

b. The trapezoidal pulse x(t) shown in figure below is applied to the differentiator defined by $y(t)=\frac{d}{dt}x(t)$ **05**

i. Find the resulting output y(t) of the differentiator

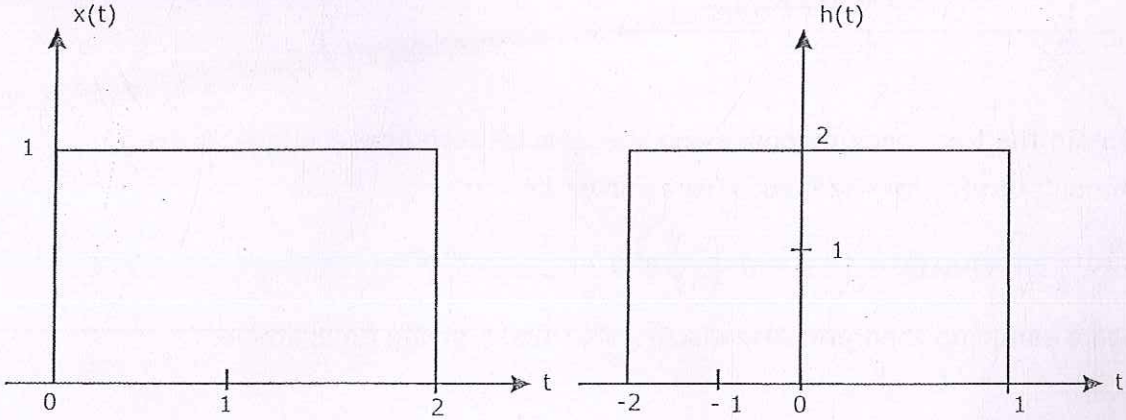
ii. Find the total energy of y(t)



c. Check whether the following syste4ms are linear, time invariant, casual memory less and stable $y(t)=x(t-2)+x(2-t)$ **05**

2. a. Prove that the output of a discrete time system can be represented as a weighted sum of shifted impulses. **05**

b. Find the response y(t) and h(t) are shown below. Use convolutional integral. **09**



3. a. Prove that if two systems are connected in series, the overall impulse response is equal to convolution of two impulse responses. 04

b. Evaluate the step response for LTI system represented by $h(t) = e^{-|t|}$ 03

c. Find the output of the system given by the differential equation. 07

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 4y(t) = \frac{d}{dt}x(t)$$

$$y(0) = 0; \left.\frac{dy(t)}{dt}\right|_{t=0}^{=1} \text{ and}$$

$$x(t) = e^{-2t}.u(t)$$

4. a. State and prove the following properties of DTFS: 08

i. Modulation

ii. Parserval's relationship

b. Find the fourier series coefficient X(k) for the signal. 06

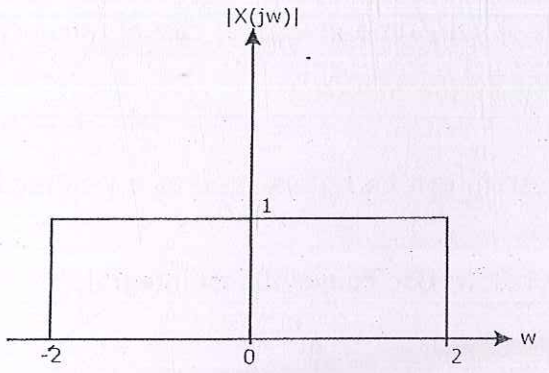
$x(t) = \sum_{m=-\infty}^{\infty}\left[\delta\left(t - \frac{1}{2}m\right) + \delta\left(t - \frac{3}{2}m\right)\right]$ and sketch the magnitude and phase spectra.

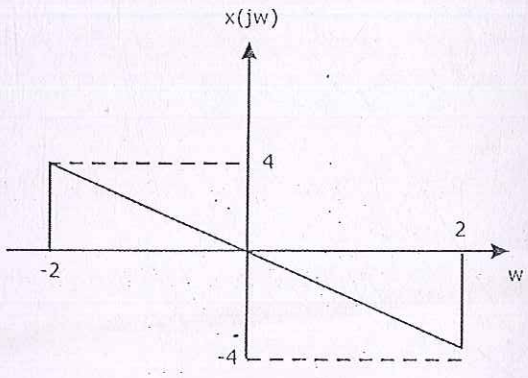
PART – B

5. a. Using the appropriate properties, find the DTFT of the following signal: 08

$$x(n) = \sin\left(\frac{\pi}{4}n\right)\left(\frac{1}{4}\right)^n u(n-1)$$

b. Determine the time domain signal corresponding to the spectra shown in figures below respectively. 06





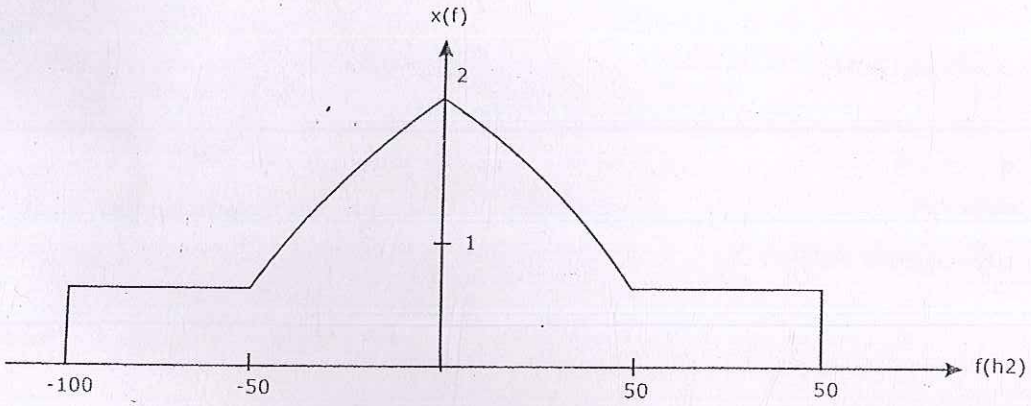
6. a. Obtain the frequency response and the impulse response of the systems a having the output y(n) for the input x(n) as given below: 05

$$x(n) = \left(\frac{1}{2}\right)^n u(n); y(n) = \frac{1}{4}\left(\frac{1}{2}\right)^n u(n) + \left(\frac{1}{4}\right)^n u(n)$$

b. State sampling theorem. Also discuss aliasing by giving an example. 05

c. A signal x(f) whose spectrum is shown in the figure is sampled at a rate of 300 04

samples /sec what is the spectrum of the sampled discrete time signal.



7. a. Determine the Z-transform of $x(n) = a^n u(n) = b^n u(-n-1)$ and find ROC. **05**

b. Explain the properties of ROC of X(z). **04**

c. State and prove initial and final value theorem of z-transform. **05**

8. a. A casual LT1 system is described by the difference equation. **07**

Y(n) = y(n-1) + y(n-2) + x(n-1).

Find the system function H(z). Plot the poles and zeros and indicate the ROC. Also determine the impulse response of the system.

b. Find the natural response of the system described by the difference equation. **07**

$$y(n) - \frac{1}{4}y(n-1) - \frac{1}{8}y(n-2) = x(n) + x(n-1) \text{ with } y(-1) = 0 \text{ and } y(-2) = 1.$$